FLATNESS OF THE LOCAL RING MAP IS ETALE LOCAL ON THE SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The spanned statement cites the chapter's own definition of a property local on the source-and-target at a point, delivered as a navigation stub.

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Allowed standard prerequisites (editorial).

- Localisations of etale ring maps are essentially of finite presentation, flat and unramified, hence faithfully flat. Algebra, Section algebra-section-etale (for the first three properties) and Algebra, Lemma algebra-lemma-local-flat-ff, invoked at proof lines 8155-8160. Makes the vertical arrows of the square of local rings faithfully flat.
- Permanence of flatness along a faithfully flat base change and along finite presentations. Algebra, Lemmas algebra-lemma-flat-permanence and algebra-lemma-flat-base-change, invoked at proof lines 8161-8173. Gives the descent direction and the first step in the converse.
- A localisation of a map between etale extensions of the same ring is flat. Algebra, Lemma algebra-lemma-map-between-etale, invoked at proof lines 8180-8182. Completes the converse direction.

Source reference. Original lemma 00HI: sources/stacks-chow/tags.txt, source lines 644–644. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 039V: sources/stacks-chow/tags.txt, source lines 4286–4286. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00U7: sources/stacks-chow/tags.txt, source lines 1101–1101. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 00U0: sources/stacks-chow/tags.txt, source lines 1094–1094. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $\mathcal{Q}$ be a property of morphisms of germs of schemes. We say $\mathcal{Q}$ is *étale local on the source-and-target* if for any commutative diagram

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

of germs with étale vertical arrows we have $\mathcal{Q}(h) \Leftrightarrow \mathcal{Q}(h')$.

Lemma 0.2. *The property of morphisms of germs*

$$\mathcal{P}((X, x) \to (S, s)) = \mathcal{O}_{S,s} \to \mathcal{O}_{X,x} \text{ is flat}$$

is étale local on the source-and-target.

Proof. Given a diagram as in Definition 0.1 we obtain the following diagram of local homomorphisms of local rings

$$\begin{array}{ccc} \mathcal{O}_{U',u'} & \longleftarrow & \mathcal{O}_{V',v'} \\ \uparrow & & \uparrow \\ \mathcal{O}_{U,u} & \longleftarrow & \mathcal{O}_{V,v} \end{array}$$

Note that the vertical arrows are localizations of étale ring maps, in particular they are essentially of finite presentation, flat, and unramified (see Algebra, Section 00U0). In particular the vertical maps are faithfully flat, see Algebra, Lemma 00HR. Now, if the upper horizontal arrow is flat, then the lower horizontal arrow is flat by an application of Algebra, Lemma 039V with $R = \mathcal{O}_{V,v}$, $S = \mathcal{O}_{U,u}$ and $M = \mathcal{O}_{U',u'}$. If the lower horizontal arrow is flat, then the ring map

$$\mathcal{O}_{V',v'} \otimes_{\mathcal{O}_{V,v}} \mathcal{O}_{U,u} \longleftarrow \mathcal{O}_{V',v'}$$

is flat by Algebra, Lemma 00HI. And the ring map

$$\mathcal{O}_{U',u'} \longleftarrow \mathcal{O}_{V',v'} \otimes_{\mathcal{O}_{V,v}} \mathcal{O}_{U,u}$$

is a localization of a map between étale ring extensions of $\mathcal{O}_{U,u}$, hence flat by Algebra, Lemma 00U7. □